ITAI 1371 – Intro To Machine Learning

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Lab 03 Individual Contribution Journal

This assignment was completed individually, so I was responsible for all the work included in my submission.

I completed the Module 03 notebook and worked through the machine learning workflow using the Wine dataset. I executed the code, reviewed the results from each stage, and worked on understanding what was happening during data preparation, splitting, training, prediction, and model evaluation instead of only running the cells.

I reviewed the exercises for supervised, unsupervised, and reinforcement learning and completed the scenario assessment identifying the correct type of machine learning for each situation. I also analyzed the results of the Logistic Regression and Decision Tree models. Logistic Regression achieved 88.9% accuracy compared with 83.3% for the Decision Tree.

For the feature selection exercise, I selected alcohol, hue, and flavanoids as the three features for my Logistic Regression model. This combination produced 94.4% accuracy, which improved the original Logistic Regression result of 88.9%. I reviewed this result and used it to better understand how feature selection can affect model performance.

I also completed an additional practice exercise using the `load_breast_cancer` dataset from Scikit-learn. For this exercise, I first explored the dataset by checking its shape, available information, feature names, and target classes. I then converted the data into a Pandas DataFrame so I could see the values in a more understandable format. I also examined the target distribution and found 357 benign samples and 212 malignant samples.

After exploring the dataset, I followed the same machine learning workflow used earlier in the lab. I divided the 569 samples into training and testing data, resulting in 455 training samples and 114 testing samples. I trained a Logistic Regression model, generated predictions using the test data, and evaluated its performance. The final model achieved 95.6% accuracy.

I also completed the written reflection and analysis sections of the notebook, including my understanding of the three learning types, analysis of the Wine classification results, a real-world application related to fashion and retail, key learnings, and questions I would like to explore further.

All coding exercises, feature experimentation, additional dataset practice, analysis, and written documentation included in my submission represent my individual contribution to this lab.